

Multimodal NLP

Topics

- Speech Recognition
 - Image Captioning
 - Multimodal Fusion
 - Visual Question Answering
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Speech Recognition

- **Speech Recognition** is the process of converting spoken words into text.
 - Input is an audio signal, and output is a sequence of words.
 - First step is to process the audio signal to extract features.
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Speech Signal Processing

- An audio signal is a 1D time series.
 - **Sampling Rate**: Number of samples per second.
 - **Bit Depth**: Number of bits per sample.
 - **Preprocessing**: Remove noise, resample, etc.
 - **Feature Extraction**: Extract features from audio signal.
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Speech Features

- **Mel-Frequency Cepstral Coefficients (MFCC)**: Popular feature for speech recognition.
 - **Short-Time Fourier Transform (STFT)**: Another popular feature.
 - These features are better suited for speech recognition than raw audio.
 - Often further features are extracted using a Convolutional Neural Network (CNN).
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Speech Recognition Models

- **Hidden Markov Models (HMM)**: See CT5120.
 - **Recurrent Neural Networks (RNN)**: As we have seen in previous lectures.
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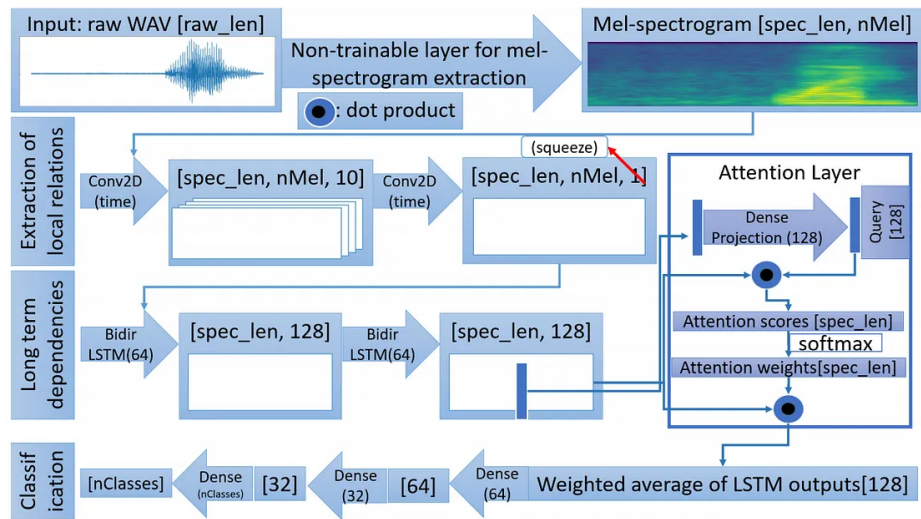


Figure 1: Speech Recognition Architecture

For more read: <https://medium.com/data-science/recognizing-speech-commands-using-recurrent-neural-networks-with-attention-c2b2ba17c837>

Image Captioning

- **Image Captioning** is the process of generating a textual description of an image.
- Input is an image, and output is a sequence of words.
- Again we need to extract features from the image

Image Features

- **Convolutional Neural Networks (CNN)**: Popular for image feature extraction.
- **Pretrained Models**: Many pretrained models are available for image feature extraction.
 - E.g. VGG, ResNet, Inception, etc.
- **Fine-Tuning**: We can fine-tune these models for our specific task.

Combining Image and Text

- Again we will use a Recurrent Neural Network (RNN) to generate the caption.
- We will include the image features using an attention mechanism.
- The attention mechanism allows the RNN to focus on different parts of the image.

Example of attention in generation

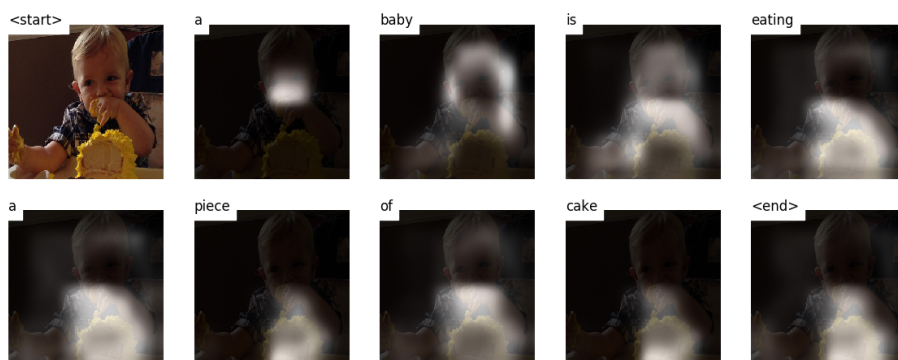


Figure 2: A baby is eating a piece of cake

Architecture of Image Captioning

For more read: <https://github.com/sgrvinod/a-PyTorch-Tutorial-to-Image-Captioning>

Visual Question Answering

- **Visual Question Answering (VQA)** is the task of answering questions about an image.
- Input is an image and a question, and output is an answer.
- Similar to image captioning but now we have a text input.

Source: Making the V in VQA Matter: Elevating the Role of Image Understanding in Visual Question Answering (CVPR 2017)

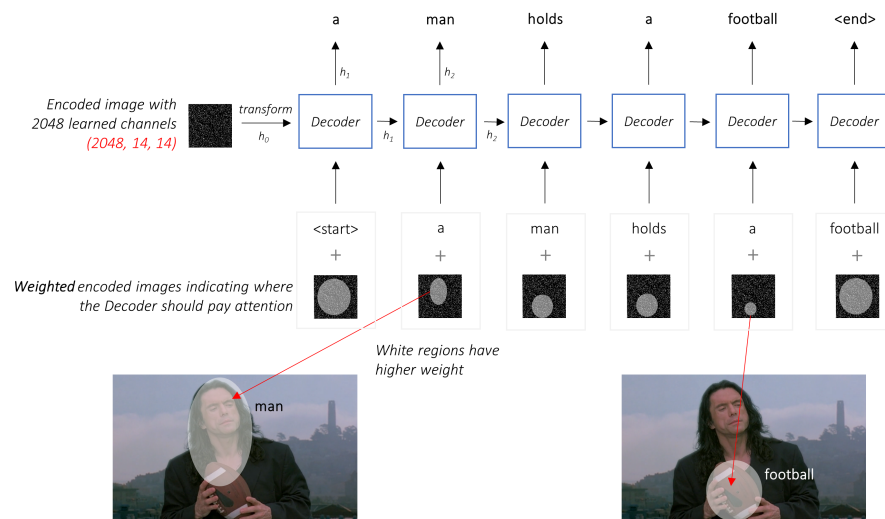


Figure 3: Image Captioning Architecture



Figure 4: Visual Question Answering

VQA Architecture

- We can use the same architecture as image captioning.
 - We will use the image features and question features to generate the answer.
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VQA Architecture

- Simple VQA architecture is to use a CNN to extract image features and an RNN to extract question features.
 - We then combine these with a dot product
 - A single layer perceptron is used to generate a **one-word** answer.
 - An RNN (or transformer) can be used to generate a **sequence** of words.
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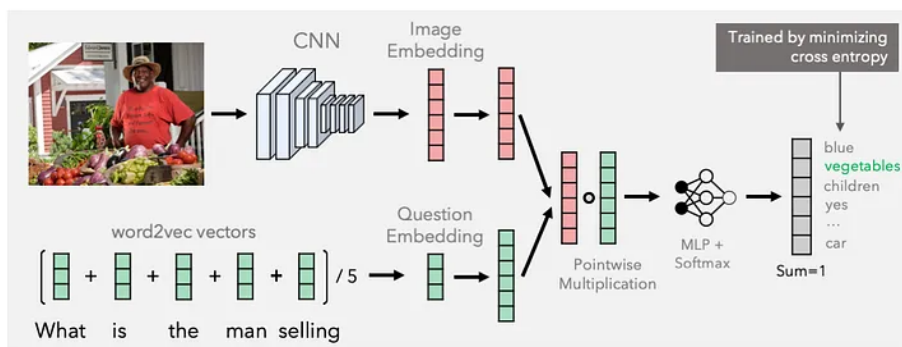


Figure 5: VQA Architecture

For more read: <https://blog.allenai.org/vanilla-vqa-adcaaaa94336>

Multimodal Fusion

- **Multimodal Fusion** is the process of combining information from different modalities.
 - **Early Fusion:** Combine features from different modalities before task-specific processing.
 - **Late Fusion:** Combine features at prediction time.
 - **Intermediate Fusion:** Combine features in task-specific processing.
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Fusion Methods

- **Concatenation:** Concatenate features from different modalities.

- **Sum/Product:** Add/multiply features from different modalities.
 - **Attention:** Use attention mechanism to combine features.
 - **Neural Network:** Use a neural network to combine features.
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Challenges

- **Heterogeneity:** Different modalities have different data types.
 - **Alignment:** Aligning features from different modalities.
 - **Interpretability:** Understanding the combined features.
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Other tasks in Multimodal NLP

- **Multimodal Machine Translation:**
 - Translation with both text and images.
 - Image can help to disambiguate words.
 - **Multimodal Sentiment Analysis:**
 - Useful for understanding sentiment in social media.
 - Image context is often vital for understanding sentiment.
 - Especially useful for sarcasm and memes.
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Figure 6: Irish State Vessel

Irish State Vessel

Other tasks in Multimodal NLP

- **Emotion Recognition**



r/funny • 10 yr. ago
daspires

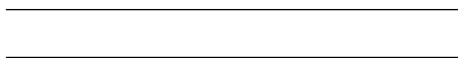


Selena Gomez, taking her music for a walk.



Figure 7: Selena Gomez taking her music for a walk

- Emotion recognition from text alone is difficult.
- Audio gives strong cues for emotion.
- **Multimodal Summarization**
 - Summarize a video with both audio and video.
 - Summarize a lecture with both slides and speech.



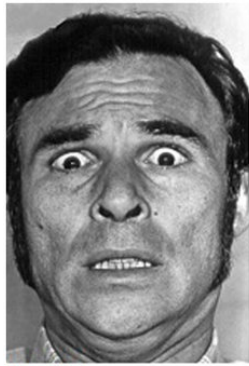
Other tasks in Multimodal NLP

- **Sign Language Recognition/Translation**
 - Recognize sign language from video.
 - Output is generally translation to spoken language (English).
 - Sign languages are not related to spoken languages
 - Irish Sign Language is not related to English and completely different from British Sign Language.



Summary

- Multimodal NLP involves combining information from different modalities.
- Speech Recognition, Image Captioning, and VQA are popular tasks in multimodal NLP.
- We can use CNNs and RNNs to extract features from different modalities.
- Features are combined by multimodal fusion methods.



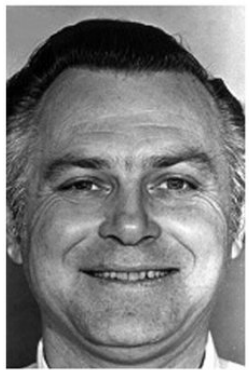
Fearful



Angry



Sad



Happy



Disgusted



Surprised

Figure 8: Emotional Faces